

Decent Coaching Classes
VIII CBSE
Mind Your Math - VI — Answer Key

Section A — MCQ

1. (b) $x^2 + 2xy + y^2$
2. (b) 1 $[76 \times 101 = 76 \times 100 + 76 \times 1]$
3. (a) $x^4 - 1/x^4$
4. (c) $(a + b)(a - b)$ $[96 \times 104 = (100 - 4)(100 + 4) = 100^2 - 4^2 = 9984]$
5. (a) $(3x + 1)^2$
6. (b) 2209 $[47^2 = (50 - 3)^2 = 2500 - 300 + 9 = 2209]$

Section B — VSA

7. $(7y + 6)^2 = 49y^2 + 84y + 36$
8. $(x^2 + 4)(x^2 + 4) = (x^2 + 4)^2 = x^4 + 8x^2 + 16$
9. $96 \times 104 = (100 - 4)(100 + 4) = 100^2 - 4^2 = 10000 - 16 = 9984$
10. $4x(5x^2 + 3x) = 20x^3 + 12x^2$

Section C — SA (any one option in each)

11. $(x + 2)(x - 2)(x^2 - 4) = (x^2 - 4)(x^2 - 4) = (x^2 - 4)^2 = x^4 - 8x^2 + 16$
OR
 $(2x - 1)(2x + 1)(4x^2 + 1) = (4x^2 - 1)(4x^2 + 1) = 16x^4 - 1$
12. $(998)^2 = (1000 - 2)^2 = 1000^2 - 2 \times 1000 \times 2 + 2^2 = 1000000 - 4000 + 4 = 996004$
OR
 $(1005)^2 = (1000 + 5)^2 = 1000^2 + 2 \times 1000 \times 5 + 5^2 = 1000000 + 10000 + 25 = 1010025$
13. $(3x + 2)^2 - (3x - 2)^2 = 4 \times (3x) \times 2 = 24x$ [using $(a+b)^2 - (a-b)^2 = 4ab$]
OR
 $(2x/7 - y/2)^2 - (2x/7 + y/2)^2 = -4 \times (2x/7) \times (y/2) = -4xy/7$ [using $(a-b)^2 - (a+b)^2 = -4ab$]

Section D — Case Study

14. (a) $34^2 - 26^2 = (34 + 26)(34 - 26) = 60 \times 8 = 480$
(b) $12x = 480 \Rightarrow x = 40$
(c) Verification : $34^2 - 26^2 = 1156 - 676 = 480 = 12 \times 40$. Hence $x = 40$ is verified.